

Morality in the Digital Age: Predictive Modelling on AITA Subreddit Posts

Kiat Hui Khang
National University of Singapore
Singapore, Singapore
e1112593@u.nus.edu

Lee Ming Xuan
National University of Singapore
Singapore, Singapore
e1103357@u.nus.edu

Jean Ong
National University of Singapore
Singapore, Singapore
e0949099@u.nus.edu

Venessa Tan
National University of Singapore
Singapore, Singapore
e1101762@u.nus.edu

Abstract—Moral and ethical judgments provide crucial guidance for individuals and societies, aligning decisions with values such as fairness, justice, and compassion. Exploring moral judgments in digital system development presents a promising avenue for enhancing ethical decision-making, fostering trust, and mitigating bias in future applications. This paper delves into the complexities of moral decision-making, leveraging the "Am I the A*****?" (AITA) subreddit as a rich source of complex, real-world moral scenarios. Through comprehensive exploration and analysis, we examine the performance of various machine learning models in discerning moral judgements. Our findings reveal that the BERT pre-trained model significantly outperforms others, achieving an F1 score of 0.90, highlighting its superior capacity to handle nuanced semantic patterns. While our study validates the effectiveness of using BERT for binary moral judgement classification, it also highlights critical areas for future research, including the enhancement of model explainability, developing nuanced moral assessments, and the need for robust testing across diverse social contexts. Our code repository is hosted at this [\[link\]](#).
(Abstract)

I. INTRODUCTION

A. Background

Moral and ethical judgments serve as compass points guiding individuals and societies toward

decisions reflective of values such as fairness, justice, and compassion. However, the inherently subjective nature of moral assessments, influenced by cultural, societal, and individual perspectives, complicates the establishment of universally applicable rules or algorithms. In essence, moral judgments are deeply entwined with our personal and collective narratives, shaped by nuanced factors beyond objective evidence.

Exploring how moral principles can be embedded into artificial intelligence algorithms and interfaces thus presents a promising avenue for not only enhancing the ethical fabric of technology but also fostering deeper reflections on morality in an increasingly digitised world.

In this paper, we try to understand if the complexities of moral judgments can be recreated using a text mining/natural language processing approach. Using machine learning models or neural networks, and leveraging datasets such as the "Am I the A*****?" (AITA) subreddit, we hope to explore whether AI systems could eventually navigate complex moral dilemmas responsibly, prioritising fairness, accountability, and regulatory compliance.

B. Objectives

Our primary objective is thus to construct a predictive model capable of discerning the moral judgement given a real-world scenario. We leverage the "Am I the A*****?" (AITA) subreddit, a dataset that provides a rich source of real-world scenarios where individuals seek moral guidance from an online community regarding their behaviour. By analysing the textual content of these posts, our model will strive to

accurately predict whether the individual in the scenario is perceived as acting morally or immorally by the community.

C. Related work

Existing studies have explored text classification within Reddit's "Am I The A*****?" (AITA) subreddit, using it as a dataset to study moral judgement and ethical decision-making expressed in narrative form. For example, Botzer et. al. [1] analysed moral judgments on Reddit, focusing on the different ethical stances users take and the linguistic markers associated with them. They highlighted the predictive capabilities of models like BERT, which achieved high accuracy in classifying moral positions based on textual content.

A seminal paper by Vosoughi and Roy (2016) [2] explored the automatic classification of the AITA posts using machine learning models, providing insights into the factors that most influence the community's verdicts. Their study demonstrated the importance of contextual and linguistic features in predicting whether users are judged as assholes or not by the community, which has important implications for understanding social norms and behaviours online.

These studies collectively demonstrate the variety of approaches that can be used to analyse the complex interactions and judgments within the AITA subreddit, providing a valuable basis for further research in text classification and ethical computing.

Building upon these foundations, our study extends the current literature by comparing traditional machine learning models with neural networks and transformer based models to discern moral judgments in real-world scenarios. We also employ other pre-processing techniques such as text summarisation to understand how they might interact with various machine models, and hence contribute to the broader discourse on the difficulty of handling human judgements.

II. DATASET

A. Data retrieval

A dataset comprising approximately 11,900 Reddit posts from the AITA subreddit was collected, along with their corresponding tags. These tags reflect the community's judgement on whether the individual described in the post is perceived as having behaved wrongly. There were some limitations in the scrapping of Reddit data as they do not allow for the filtering of posts by date. Hence, the dataset is scrapped based on the top 1000 submissions of different categories within the subreddit, using different sorting methods like 'top', 'hot' and 'rising'. Features used for machine learning would be purely derived from the content in the post, and does not include other variables like title, author and date of post.

The reddit posts were filtered down to three primary tags - 'Not the A-hole', 'Asshole', and 'No A-holes here', as well as posts which had 50 or more 'tokens'. Other tags were ambiguous in terms of whether they were morally right or wrong, and hence removed from the dataset. Posts which were less than 50 tokens were partially taken down (i.e. the author removed their content) and hence does not have sufficient information for the machine learning model to learn from.

B. Data labelling

Thereafter, posts tagged as 'Not the A-hole' and 'No A-holes here' were further categorised as instances

where the behaviour described was not considered morally wrong. Conversely, posts tagged as 'Asshole' were identified as cases where moral wrongdoing was perceived.

This strategy reshaped our project into a binary classification task. This simplification allowed for a more targeted approach in training our models, where the primary objective became effectively distinguishing between morally wrong and not morally wrong actions as perceived by the community.

C. Exploratory Analysis

An Latent Dirichlet Allocation (LDA) model was applied to uncover prevalent themes within the dataset, revealing that the majority of discussions could be categorised into three main contexts: family, work, or school, with frequent topics including trips, meals, and financial dealings. This thematic distribution reflects the everyday moral dilemmas faced by individuals on interpersonal issues.

D. Addressing Dataset Imbalance

Initially, our dataset exhibited a significant imbalance, with 10,134 entries labelled as not morally wrong ('0') and only 1,782 as morally wrong ('1'). This imbalance can bias our models towards the majority class, resulting in poor generalisation performance on the minority class. To rectify this imbalance, we augmented the dataset with 2,462 entries sourced from a publicly available dataset [3], as well as 3,536 entries from the ETHICS dataset [4] which had been originally sourced from the AITA subreddit as well. These additions were carefully selected to match the context and source of our initial dataset.

E. Train-test split

The final dataset was divided into training and testing sets. The training set consists of 10,872 samples, which are balanced between 5,134 samples labelled as 'not morally wrong' and 5,738 labelled as 'morally wrong'. Conversely, the testing set includes a total of 6,000 samples, distributed between 5,000 'not morally wrong' and 1,000 'morally wrong' samples. While the test dataset exhibits a greater imbalance, potentially increasing the likelihood of false positives, we prioritised a balanced training dataset for our study. Depending on the model's requirements, the training data was then further split for validation or used in k-fold cross-validation.

III. METHODOLOGY

This section outlines the approaches and techniques applied to the prepared dataset.

A. Pre-processing techniques

In the preprocessing stage, various techniques were explored in order to enhance the quality of the text data while maintaining its versatility for different text representation methods. Depending on the models, text normalisation techniques such as case folding, stopword removal and lemmatization were selectively applied.

B. Summarisation techniques

Additionally, given the varied length and detailed nature of Reddit posts in the "Am I the A*****?" subreddit, we explored text summarisation techniques to distil essential content and remove noise for selected models. We applied two summarization approaches. Firstly, leveraging the capabilities of the pre-trained GPT-2 model, we experimented with generating concise

summaries of the posts. Secondly, we adapted the PageRank algorithm to evaluate the centrality and relevance of sentences within each post. We strategically chose to summarise each post into five sentences, minimising the need for padding sentences as less than 1% of posts were shorter than 5 sentences.

C. Text representation approaches

Our exploration encompassed a variety of text representation techniques tailored to the specific requirements of each model. These techniques aimed to transform raw textual data into numerical formats suitable for processing by machine learning and deep learning models, and include TF-IDF, handcrafted features, TensorFlow Tokenizer Sequences, and BERT embeddings. These approaches were applied to the original and summarised versions of the data.

C. Model Selection

The methodologies implemented in our study can be broadly categorised into two primary groups: traditional machine learning (ML) approaches and deep learning (DL) strategies:

- **Traditional Machine Learning Approaches:** Naive Bayes and Logistic Regression models were explored to provide a foundational comparison against more sophisticated methods, and to explore whether TF-IDF, handcrafted features, and sentence-based approaches could provide sufficient information for a model to make moral judgements.
- **Deep Learning Approaches:** Deep learning architectures such as Bi-directional Long Short-Term Memory (Bi-LSTM) and Convolutional Neural Networks (CNN) were employed. Additionally, a fine-tuned BERT model, a type of Transformer architecture leveraging contextual embeddings for text classification, was explored.

These approaches were chosen to explore different aspects of machine learning and natural language processing to enrich our comparative analysis.

D. Evaluation Metric

To gauge model performance, F1 score was selected as the primary metric, as it offers insight into its balance between precision and recall, particularly important for classification tasks.

In the context of moral judgments, precision reflects the model's accuracy in identifying posts as morally right or wrong—ensuring that each label assigned by the model corresponds accurately to the true nature of the post. This is crucial because incorrect labels could misguide users and foster distrust in the system's judgement.

Recall, on the other hand, measures how much the model successfully identifies most of the morally pertinent scenarios (true positives). Missing out on these (false negatives) could mean failing to flag morally questionable content, which is equally problematic as it suggests a lack of discernment in the model. F1 was therefore selected as the determining metric. Thus, using the F1 score as the evaluation metric ensures that our model is not only accurate in its classifications but also comprehensive in detecting all pertinent cases, thus maintaining reliability and trust in its moral judgments.

IV. TRADITIONAL MACHINE LEARNING APPROACHES

For traditional machine learning models such as Naive Bayes (NB) and Logistic Regression (LR), we

expanded contractions, and performed case folding, stopword removal and lemmatization for text preprocessing with spaCy. We utilised TF-IDF representation to convert text documents into vector models, emphasising unique words in a document.

TABLE I. PERFORMANCE OF TRADITIONAL ML APPROACHES

Model	Dataset	Representation	F1 score (Test)
NB	AITA	Lexicon features	0.29
	AITA	TF-IDF	0.48
LR	AITA	Lexicon features	0.30
	AITA	TF-IDF	0.50
2-stage NB	AITA	TF-IDF	0.29
2-stage LR	AITA	TF-IDF	0.39

Fig. 1. Test F1 scores for each model. The best results are in bold.

A. Lexicon-based approach

Firstly, we explored utilising handcrafted features based on a moral lexicon. The rationale behind using a moral lexicon-based approach in our study was to systematically incorporate predefined moral values into the model's analysis, enabling a nuanced assessment of textual content against established moral codes. We incorporated the MoralStrength lexicon [5], a comprehensive resource consisting of approximately 1,000 lemmas that have been evaluated across moral dimensions derived from the Moral Foundations Theory (MFT) [6], a well established theory in the social and psychological sciences.. The numeric scores for each of the 5 moral valence, as well as the average score across all 5 moral valence, were utilised as handcrafted features. Each Reddit post was scored using the mean of the moral valence scores of the individual words of the post.

Using these handcrafted features, both Naive Bayes and Logistic Regression models were trained, with similar F1 scores of 0.29 and 0.30 respectively. This could be attributed to these features' inability to capture contextual information from the text, or the limited vocabulary of the lexicon, which resulted in the presence of null values for the features in a significant portion of the dataset. Even after adjustments to treat null values as neutral values, the models still performed poorly, suggesting that the moral lexicon-based features might not be robust or informative enough for the task. These results underscore the challenge in feature extraction and selection, and the critical need for models that can better generalise from training data to unseen data, particularly in complex text classification tasks involving diverse linguistic inputs such as lengthy Reddit posts.

B. Traditional Machine Learning

We then applied the Logistic Regression and Naive Bayes algorithms to the entirety of the original text data by vectorising the text data using TF-IDF. This improved the performance of the models, with both NB and LR models achieving respectable cross-validation scores of 0.74 and 0.72 respectively. However, their performance on the test dataset was markedly lower, with F1 scores of 0.48 and 0.50. This significant drop in performance on unseen data suggests a substantial case of overfitting. The traditional models, despite their probabilistic efficiency and simplicity in handling text data, might be overly sensitive to noise or overly fitted to specific distributions in the training data, which undermines their ability to generalise to new data.

C. 2-step traditional machine learning

Enhancing the complexity of traditional approaches, a two-step model was explored, involving the segmentation of each Reddit post into its constituent sentences, dissecting textual content into morally evaluative units. These sentences were then individually analysed using traditional machine learning techniques, specifically Logistic Regression and Naive Bayes, to label each sentence based on its moral content. The labels from these sentences were subsequently aggregated and passed through another model to predict a final moral verdict for the entire post. The hypothesis behind such an approach is that morality could be focused on some of the key sentences in the passage, and we could assign different levels of importance on different parts of the passage based on the outcome of the second regression.

The two-stage Logistic Regression model achieved the best F1 score of 0.39. This score is significantly lower than just using a traditional approach.

Upon closer inspection, we discovered that the second regression in our approach functioned essentially as a majority vote among sentences, disregarding the contextual significance of their placement within the overall passage. This finding contradicts our initial hypothesis, which suggested that the order of sentences might influence the moral judgement of the text or that a post could be evaluated without full context. The variation in writing styles among different Reddit users may contribute to this result; some may present key information at the beginning, while others might place it towards the end.

This observation also reaffirms that knowing only a specific segment of a passage is insufficient for accurately determining its moral alignment, which aligns with intuitive reasoning. The second model's inability to incorporate additional contextual information effectively degraded its performance compared to using a single traditional machine learning model that evaluates the entire passage. Thus, providing the model with limited data (only five sentences rather than the full text) inherently restricts its understanding and leads to poorer outcomes.

As per the learnings from just using traditional machine learning models, the use of word vectors without additional context does not provide sufficient information for the model to determine morality.

In addition, a significant limitation of this methodology involves the assignment of standard labels to individual sentences within the training and test data based on the overall or final label of the complete text. This approach presupposes that each sentence inherently reflects the moral judgement or classification of the entire text, leading to the inevitable introduction of incorrect classifications. While we have also tried to explore other sources of data, to help label individual sentences, none provided a sufficiently high score to be used in this approach.

V. DEEP-LEARNING APPROACHES

Our implementation of deep learning architectures included exploring Bi-LSTMs, CNNs, as well as fine-tuning a pre-trained BERT model. To maximise the retention of nuanced information crucial for accurately interpreting moral and ethical contexts, texts were not subjected to preprocessing before model input. This approach ensures that the full richness of the data,

including subtle linguistic features, is available for analysis.

For all models in this section, we assess their performance on the 3 variations of the dataset—Original (ORIG), GPT-2 Summarised (GPT2-S) and PageRank Summarised (PR-S).

A. Recurrent and Convolutional Neural Networks

To determine the optimal input parameters, we tailored the sequence length and vocabulary size to suit the specific needs of each model and the data variation being used (summarised or full text). The maximum sequence length was selected according to the 95th percentile of sequence lengths within the dataset to capture most of the relevant information without truncating critical content.

TensorFlow's tokenizer was used to assign a unique integer to each word and convert texts into sequences of these integers, each representing a word in the corpus dictionary. This method maintains the sequential nature of the text for models such as Bi-LSTMs and CNNs.

The Bi-LSTM architecture was explored due to its ability to capture both past context and future context, which is pivotal for understanding the full context of each post. The model starts with an embedding layer, followed by a 32-unit bidirectional LSTM layer. The output from the LSTM layer passes through a fully connected dense layer with 64 units and ReLU activation to introduce non-linearity and aid in learning complex patterns. A dropout layer with a rate of 0.1 is used to prevent overfitting by randomly setting input units to 0 during training. The final output layer is a single neuron with a sigmoid activation function, designed to output the probability of the input being classified as morally wrong. A variation of the model included integrating a multi-head self-attention layer with four heads, together with an additional bi-LSTM layer.

The CNN Model utilises convolutional layers to capture local feature patterns within the text, which can be indicative of moral sentiments. The architecture comprises an embedding layer followed by a multi-head self-attention layer. Three convolutional layers, each paired with a global max pooling layer, are applied consecutively, with decreasing filter sizes of 10, 5, and 3 to capture progressively finer details. The outputs from all pooling layers are concatenated to form a comprehensive feature vector, which is then processed by a dense layer and a sigmoid output layer. This model is particularly adept at processing inputs where local contextual cues within a text are critical for classification. A variation of the model included adding a MultiHeadAttention layer.

Model training parameters were standardised across experiments for consistency. Each model was trained with a binary cross-entropy loss function and an Adam optimizer, across 10 epochs with a batch size of 128.

TABLE II. PERFORMANCE OF NEURAL NETWORKS

Model	ORIG	GPT2-S	PR-S
Bi-LSTM	0.35	0.56	0.38
Bi-LSTM with Attention	0.43	0.44	0.38
CNN	0.41	0.47	0.39
CNN with Attention	0.57	0.47	0.39

Fig. 2. Test F1 scores for each model variant and summarised version of the text. The best results are in bold.

In our study, the CNN demonstrated slightly superior performance (0.57) compared to the Bi-LSTM when trained on original data. This superior performance of the CNN can largely be attributed to its architectural efficiency, which excels in extracting local and positionally relevant features from text. Such features are crucial for accurately interpreting moral judgments from our dataset. Additionally, CNNs benefit from parallel processing, which not only reduces training times but also enhances the handling of large datasets with complex features without being hampered by the sequential processing constraints seen in Bi-LSTMs.

Using the original dataset, the inclusion of an attention mechanism in both CNN and Bi-LSTM architectures generally improved performance on the original texts. This suggests that attention mechanisms help the models to better focus on the most informative parts of the texts, which is crucial when dealing with full-length posts where relevant information might be embedded in a larger context. However, for GPT-2 summarised texts, the simpler, base models without attention performed better and reached comparable scores (0.56) to the CNN with Attention model trained on original data (0.57), indicating that the condensed and focused nature of the summarised texts might make the additional complexity of an attention mechanism less beneficial or even redundant.

Interestingly, when using PageRank-summarised texts, both Bi-LSTM and CNN models did not show improved performance, and the incorporation of attention mechanisms did not significantly enhance model effectiveness. This could suggest that the type of summarization and its alignment with the model's capabilities are crucial. PageRank summaries, which might focus on different aspects of the text compared to GPT-2, possibly do not align as well with what the models, particularly those with attention, are optimised to process. This disparity underscores the importance of matching the model architecture and its complexities with the characteristics of the text representation used, highlighting that more sophisticated models do not always equate to better performance, especially when the textual input is already optimised or simplified.

Despite efforts to optimise their configurations, both CNN and Bi-LSTM architectures ultimately did not achieve satisfactory F1 scores, particularly struggling with low precision characterised by a significant number of false positives, as illustrated in the confusion matrices below (Fig. 2).

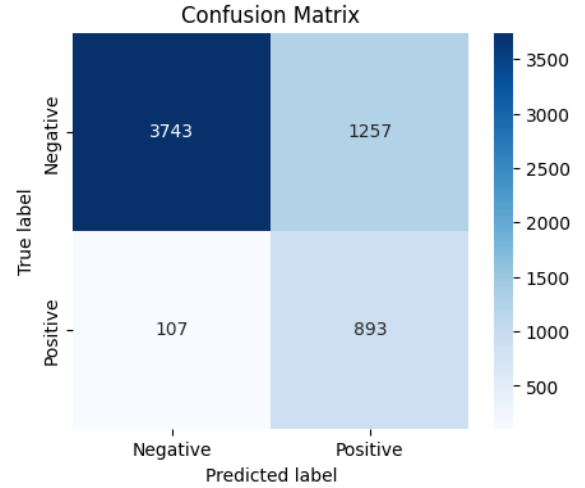
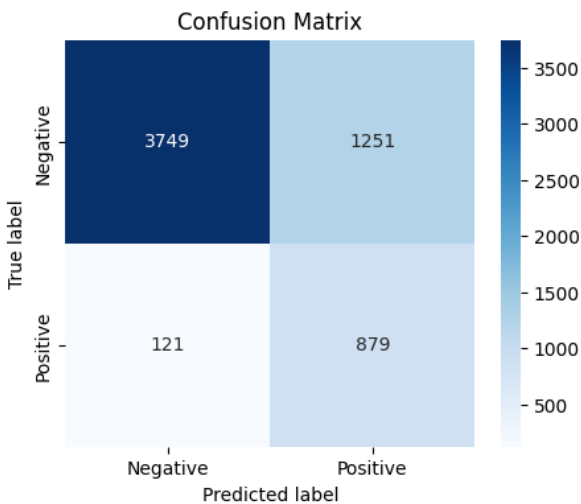


Fig. 3. Confusion matrices for the best performing Bi-LSTM and CNN models respectively.

The substantial number of false positives indicates a fundamental issue in these models' ability to distinguish accurately between classes. Potential reasons for this limitation include the inherent nature of Bi-LSTMs and CNNs. Although Bi-LSTMs are designed to capture long-range dependencies better than unidirectional RNNs, they may still struggle with extremely long texts or complex narrative structures due to an insufficient context window. Furthermore, Bi-LSTMs are prone to overfitting on complex and varied data like moral judgments, where they might adapt too specifically to training patterns, hindering generalisation to new data. CNNs, adept at identifying local patterns, may overlook the broader context crucial in understanding the complex narrative structures typical in datasets involving moral judgments.

B. Fine-tuning Pre-trained Models

The paper "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding" [7] highlights that the pre-trained BERT model can be effectively fine-tuned by adding just one additional output layer to achieve state-of-the-art (SOTA) results on eleven natural language processing tasks. In this section, we explore the potential of a fine-tuned BERT model for our classification task. We aimed to leverage BERT's comprehensive language understanding without training a model from scratch.

We chose to utilise the smaller uncased $BERT_{BASE}$ model ($L=12$, $H=768$, $A=12$, Total Parameters=110M), which does not differentiate between uppercase and lowercase letters. Our rationale was that the uncased model, potentially capturing less information, would provide a baseline for the simplest form of the model's capabilities. Our fine-tuned model involves appending a single dense layer, consisting of two output units with a 'softmax' activation function. This allows us to interpret the network's output as confidence scores for class 0 and class 1. More specifically, we pass the pooled output from the pretrained model to the dense layer, as we only require one hidden state per sentence for our classification task. We denote our fine-tuned model as BERT-FT.

Given that BERT's contextual capacity is limited to 512 tokens, and given that 36.72% of our training set and 55.33% of our test set exceeded 512 tokens respectively, we hypothesised that our model would learn enough from the train set, and score at least 0.45

on the final accuracy. Additionally, we also hypothesised that BERT-FT’s performance would outperform models trained from scratch. With regards to summarised data, we hypothesised that it might diminish the contextualised representation’s richness captured by the encoder. Considering that BERT has been trained on extensive datasets, the model’s learned representations could benefit more significantly from the original dataset, leading to better performance.

For BERT-FT, it was constrained to a maximum of 10 epochs. We retained the model state corresponding to the minimum value of the used loss function on the validation dataset for evaluation purposes. Keeping our model constant and varying only the datasets – namely ORIG, GPT2-S and PR-S, our results are shown in Table III.

TABLE III. PERFORMANCE OF BERT-FT

Dataset	Precision	Recall	F1
ORIG	0.81	1.00	0.90
GPT2-S	0.57	0.98	0.72
PR-S	0.45	0.82	0.58

Fig. 4. BERT-FT’s Performance on original and summarised datasets

Overall, the results confirmed our hypothesis, with the best performance achieved with the original dataset. Between the summarised datasets, the GPT2-S-trained model achieved a better performance. The initial results suggest that critical nuances may be lost during the summarization process, which could diminish the effectiveness of subsequent moral judgement analysis by machine learning models. Notably, from the best to worst performing, there was an increasing trend for both false positives (FP) and false negatives (FN).

TABLE IV. CONFUSION MATRIX (ORIG)

True Label	Predicted Label	
	0	1
0	4769	231
1	0	1000

Fig. 5. BERT-FT’s Confusion Matrix on original data (ORIG)

From Table IV, we identified 231 instances classified as false positives. Within these, 56.28% had input sequences with token counts exceeding 512, while 43.72% did not exceed this threshold. Notably, of the sequences that did exceed, a significant majority (83.85%) had token counts ranging between 1 and 300. This pattern suggests that key information likely embedded within the first 300 tokens was lost due to truncation, which, coupled with the model’s inability to capture nuances in the input sequences, adversely affected its accuracy and confidence. Despite the relatively complex architecture of BERT-FT, it was anticipated that the model might not fully comprehend the knowledge from input texts to make a confident judgement. It is expected that these issues would worsen with summarisation.

TABLE V. CONFUSION MATRIX (GPT2-S)

True Label	Predicted Label	
	0	1
0	4257	743
1	18	982

Fig. 6. BERT-FT’s Confusion Matrix on summarised data (GPT2-S)

TABLE VI. CONFUSION MATRIX (PR-S)

True Label	Predicted Label	
	0	1
0	4006	994
1	184	816

Fig. 7. BERT-FT’s Confusion Matrix on summarised data (PR-S)

C. Analysing Fine-Tuned Models on Test Examples

For a more detailed analysis, we selected two examples from the test set, each with true labels of 1 and 0, and examined the probabilities of each of the three models assigned to both classes. When assessing sentences from GPT2-S, PR-S, and ORIG, we use their corresponding models: GPT2-S-trained, PR-S-trained, and ORIG-trained, respectively.

TABLE VII. CLASS PROBABILITIES ON 5002TH EXAMPLE

Dataset	Prob (0)	Prob (1)	True Label
ORIG	0.0003	0.9997	1
GPT2-S	0.0006	0.9994	1
PR-S	0.0012	0.9988	1

Fig. 8. Class probabilities on 5002th example from test set

From the 5002th example in the test set, the GPT2-S summarised version of 62 words from the original 247 words is ***“AITA for making out with this dude’s ex in front of him? But last night I’m at a bar, and both Mark and Jia are there. I tend to think that PDA of that kind is a bit trashy so after a few seconds I get up with her and we go outside, but I can see that Mark has been watching the entire time.”*** It was interesting that even though 74.90% of the words were removed, the probability of the correct class only fell by 0.0003, as shown in Table VII. Considering that each model was trained on distinct datasets (both summarised and non-summarized), in this instance, the summarised data from the train set used for both PR-S and GPT2-S-trained models provided sufficient exposure for the models to accurately and confidently classify this example. We bolded the sentence that we suspected influenced the model’s overall predictions towards class 1. We denote this sentence as S1.

The same example in PR-S was summarised to 93 words, ***“But last night I am at a bar, and both Mark and Jia are there. Jia’s with her friends as well but after a time comes over to my table and sits next to me, starts chatting me up. I play rugby with the guy in question (let us say, “Mark”) but I do not usually hang out with him outside of matches and practice. He broke up with a woman (Jia) that I am rather attracted to last week. AITA for making out with this dude’s ex in front of him?”***

To determine which sentence had the highest probability for class 1, we ran a loop testing each sentence with the relevant model. S1 consistently showed the highest probability across all models, with scores of 0.9996, 0.9213, and 0.5175 for GPT2-S, PR-S, and ORIG models respectively.

From our observations, it’s notable that sentences containing ‘but’, which typically introduce a form of justification, achieved lower probabilities for class 1. For example, the first two sentences from the PR-S example had probabilities of 0.3950 and 0.4035. Even in high-scoring GPT2-S examples, sentences with ‘but’ had the lowest class 1 probabilities. This pattern was

consistent in the ORIG-trained model as well. To confirm this, the phrase “*but i was drunk*” was appended at the back of S1 and fed into the models again. We denote this sentence as S2. This brought the probability of class 1 from 0.9996 and 0.9213, to 0.9933 and 0.9111 respectively. In the ORIG-trained model, the probability of class 1 for S2 increased compared to S1, from 0.5175 to 0.5729. This increase can be attributed to the model’s training on a more extensive dataset, where adding the statement did not lead to a more forgiving assessment.

This suggests that models trained on summarised data may recognize simpler justifications and become more forgiving but face overall limitations in judging a situation with topics involving relationships, which often present ambiguities. In our analysis, the primary reason for classifying S1 as class 1 can be attributed to a widely recognized social norm that advises against involvement with a friend’s ex-partner. However, this interpretation could vary based on the perspective that in matters of love, conventional rules may not always apply, suggesting that these social norms might not be universally binding. The ORIG-trained model reflects this ambiguity by showing closely aligned probabilities for class 0 and class 1. In contrast, the models trained on summarised data displayed larger gaps in probabilities for S2, indicating a less nuanced understanding of complex scenarios. This shows that the ORIG-trained model’s exposure to complex topics likely enhanced its understanding of inherent ambiguities, reflecting the complexity of its training data.

TABLE VIII. CLASS PROBABILITIES ON 1157TH EXAMPLE

Dataset	Prob (0)	Prob (1)	True Label
ORIG	0.9038	0.0961	0
GPT2-S	0.9920	0.0080	0
PR-S	0.7377	0.2623	0

Fig. 9. Class probabilities on 1157th example from test set

From the 1157th example in the test set, GPT2-S summarised version of 116 words from the original 584 words is “*I (F44) live with my husband (M47), we have two adult children who don’t live with us anymore. Two years ago MIL was diagnosed with Alzheimer’s disease. After I finish work, I take care of the house and MIL. I don’t have enough patience, I don’t like when other people depend on me, I am nervous when I suddenly have to adjust my plans. I do my best to not to show it but the situation really wears me off. My husband has mixed feelings about it, which is understandable as she is his mother, but he says that given that I’m the one mostly taking care of her, the final decision is mine to make.*” In our analysis, we noted that key information was missing in the summarised versions of the story. Specifically, from the above, the writer’s central dilemma about whether to send her mother-in-law (MIL) to a specialised care facility was not included, omitting a critical aspect of the narrative.

This example was more complex than the previous, as it involved multidimensional issues such as caregiving, family dynamics, work-life balance, ethical decisions related to elder care, variable family responses, emotional depth and personal reflections. Even though all models correctly classified the example with high confidence as seen from Table VIII, their success appeared to be due to their focus on a limited aspect of the narrative. For example, the GPT2-S-trained model exhibited a high probability of 0.9313 for class 0 on the sentence “*I do my best to not*

to show it but the situation really wears me off.” Meanwhile, the PR-S-trained model showed a high probability of 0.7500 for class 0 on the sentence “*I do not want to let a stranger into my house AND I do not want to be a caretaker at all and in this configuration I would still have to take care of MIL after work.*” The ORIG-trained model yielded a high probability of 0.8563 for class 0 on the sentence “*As for me, I usually work 8-10 h/day, mostly from home but sometimes I have to be in the office. After I finish work, I take care of the house and MIL.*”

This suggests that the models might be learning about the writer’s responsibilities, reflecting a common perception on Reddit where posters often discuss their burdens or express feelings of being misunderstood. To confirm this, we run this sentence which we denote as S3, on all models. “*As for me, I usually work 8-10 h/day, mostly from home but sometimes I have to be in the office. After I finish work, I take care of the house and MIL.*” All models yielded high probabilities on class 0 for S2. Models trained on ORIG, GPT2-S and PR-S respectively yielded probabilities of 0.8608, 0.9217 and 0.6980 for class 0 respectively. An interesting observation was with the sentence “*Two years ago MIL was diagnosed with Alzheimer’s disease.*” that the ORIG-trained model assigned a strong class 0 probability of 0.7962. In our analysis we find this should be labelled as class 0 as well. With the same sentence, the GPT2-S-trained model assigned a strong class 1 probability of 0.9749 for the wrong class, leading to a false positive while the PR-S-trained model assigned a strong class 0 probability of 0.6884. Through further analysis, we discovered that the features learned from the summarised data seem to represent only a subset of the more comprehensive features available in the ORIG dataset, and the misclassification by the GPT2-S-trained model highlighted a common limitation among all models in handling complex topics. These models may struggle to fully grasp the nuanced contexts and emotional subtleties within stories, potentially leading to inaccuracies like false positives.

VI. SUMMARY OF RESULTS AND DISCUSSION

Our comprehensive evaluation of various models reveals significant differences in performance across the spectrum of approaches we implemented. From the summarised results below, it is evident that although BERT-FT had limitations, it significantly outperformed all other models:

TABLE IX. PERFORMANCE OF ALL MODELS ON ORIGINAL DATASET

Model	Representation	F1 score (test)
LR	Lexicon features	0.30
LR	TF-IDF	0.50
2-stage LR	TF-IDF	0.39
Bi-LSTM	Sequences	0.56
CNN	Sequences	0.57
BERT-FT	BERT embeddings	0.90

Fig. 10. Summary of best F1 scores for each approach.

Lexicon-based models attempt to encode moral principles directly into the decision-making process. However, our experiments have shown that these models do not inherently learn or internalise these principles. Instead, they function more as classifiers that match patterns without a deep understanding of underlying moral reasons. They also struggle with the

context-dependent nature of language, where the moral value of words can shift dramatically based on context, usage, and narrative structure.

In exploring traditional machine learning models like NB and LR for text classification, we noted that these approaches face limitations in processing complex texts such as lengthy Reddit posts, which often contain nuanced language and syntax. Due to their reliance on linear assumptions and simpler text representations like TF-IDF, these models struggle to capture deeper semantic patterns, limiting their effectiveness in scenarios with sophisticated linguistic content. Therefore, while useful for benchmarks and preliminary analyses, it is clear that traditional machine learning models do not offer the best performance for complex text classification tasks.

Bi-LSTMs and CNNs, despite their popularity in various text processing tasks, face significant challenges when applied to the nuanced domain of moral judgments, particularly in the context of Reddit posts. Reddit posts, often characterised by their informal language, varied topics, and complex narrative structures, present unique challenges that test the limits of these neural architectures. The relative underperformance of Bi-LSTMs and CNNs could be attributed to a few factors. Firstly, the dataset size might not have been sufficient to train these deep learning models effectively without leading to overfitting. In contrast, BERT benefits from its pre-trained background, allowing it to excel even on smaller, domain-specific datasets. Secondly, the architectural design of the neural networks may not have been sufficiently complex to capture the intricate semantic and syntactic nuances necessary for analysing complex moral judgement texts.

BERT-FT's superior performance, with an F1 score of 0.90, underscores its robustness in handling complex semantic patterns. BERT's ability to leverage vast amounts of pre-training on diverse language data allows it to excel in tasks requiring deep semantic understanding, unlike models trained from scratch on the dataset at hand. Crucially, BERT's advanced text representation techniques offer a much richer and more context-aware understanding of text semantics. This advanced capability is likely why BERT significantly outperformed other models in our tests. Its robust performance, even on smaller datasets, underscores the critical role of sophisticated, context-aware embeddings in achieving high accuracy in specialised NLP tasks. BERT's ability to leverage vast amounts of pre-training on diverse language data allows it to excel in tasks requiring deep semantic understanding, unlike models trained from scratch on the dataset at hand.

VII. CONCLUSION

This study has laid the groundwork for developing a predictive model capable of discerning moral judgments within the context of the "Am I the A*****?" (AITA) subreddit.

However, despite BERT's impressive performance, it is clear that having a high-performing classifier does not fully resolve the challenges associated with this task. We recognize that even if a machine learning model effectively predicts morality using F1-scores for justification, it fails to consider the cultural and emotional perspectives that influence human judgments. Culturally, interpretations of morality can vary widely; individuals might disagree with standardised labels

provided in datasets like those from Reddit, based on their personal or societal values.

Furthermore, in practical applications where morality is assessed, the ability to explain decisions is crucial. Typically, when someone is judged for wrongdoing, the judgement is supported by clear evidence and reasoning. However, our current model determines whether an action or passage is morally right or wrong without providing the underlying reasons for these conclusions. This is a significant limitation because it lacks transparency, which is essential for trust and accountability in decision-making processes.

These issues point toward the need for further exploration to enhance both the scope and depth of our research.

One promising avenue is the development of enriched datasets that include not only examples of moral judgments but also contextual explanations for why specific actions are deemed right or wrong. Specifically, for the AITA dataset, incorporating user comments could serve as explanatory data, enhancing the model's understanding of community-driven moral reasoning. These comments often unveil the rationale behind users' judgments, providing diverse perspectives that are invaluable for training more nuanced models.

As an extension of the above, there is potential to expand beyond binary classifications to include multi-class systems or incorporate aspect-based moral analysis. Such an approach would allow for a more nuanced understanding of moral judgments, potentially quantifying degrees of responsibility or different aspects of moral behaviour. This shift would not only improve the granularity of the assessments but also align more closely with the complex nature of human moral reasoning, where actions are rarely seen as merely right or wrong.

Next, to improve model explainability, future research should focus on developing techniques that allow a model to not only make moral judgments but also to articulate what aspects of a situation influenced its decisions. By doing so, the model would not just offer conclusions but also provide the rationale behind them, much like a human would. Investigating hybrid models that integrate lexicon-based approaches with advanced language models like BERT could offer a solution, bridge the gap between high performance and human-like reasoning and explainability. Such advancements would not only enhance the model's utility but also foster greater trust and acceptance of automated moral assessments.

Finally, it is crucial to test the adaptability and effectiveness of these models across various platforms and contexts by conducting comprehensive evaluations using unseen textual data from diverse sources. This would ensure that the models are robust and versatile enough to handle different discourse styles and moral contexts.

These future directions promise to enhance our understanding of moral decision-making processes and advance the development of AI systems capable of navigating complex ethical dilemmas in diverse online communities. Ultimately, the goal is to create AI that not only evaluates morality but does so with a clarity and depth that mirrors human ethical considerations.

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